# FROM TOOLS TO TEAMMATES

## THE RISE OF AI AGENTS IN SOFTWARE DEVELOPMENT

By Ashadeepa Debnath

2026

JS COMMUNITY

# Why Today's Tools Aren't Enough

## THE SHIFT

- Modern software is a living city of services, APIs, pipelines, and teams.
- The challenge is not only building components but coordinating them reliably at scale.

## TOOLS ARE REACTIVE

- Autocomplete, linters, and copilots help developers move faster — but they wait for prompts.
- They reduce effort without owning goals, context, or outcomes.

## THE HUMAN BOTTLENECK

- When every decision still routes through a person, delivery slows.
- The system can be powerful, but the orchestration layer remains manual and fragile.

JS COMMUNITY

Systems are powerful, but orchestration is still manual and fragile

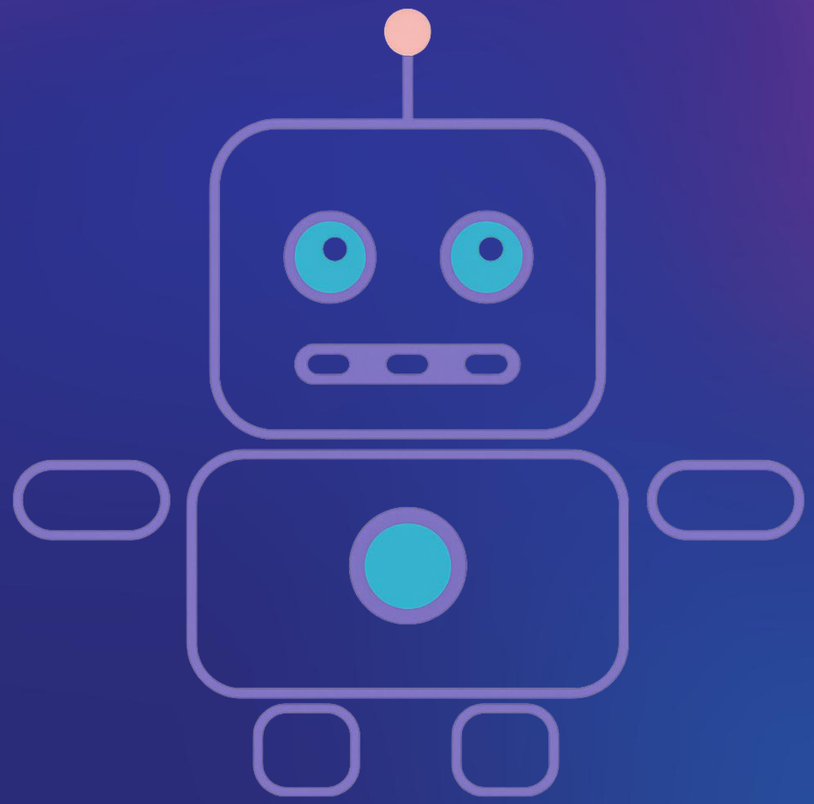
# Who Am I

- **AI-enabled architect** and **developer at** ❤️ with a mission to build scalable, secure, future-ready platforms across frontend, backend, and cloud.
- A silent force of innovation with **3 US-filed patents2- already granted patents**.
- Bringing ~**13 years of IT experience** and the precision of a top-tier operative to every mission.

Technologist, Passionate, Creative!

JS COMMUNITY

# The Shift Nobody Noticed



## Era 1 — Autocomplete (2022)

Copilot, code suggestions. Stateless. You invoke, it predicts.
Same mental model as a very smart search.
No goals, no context, no outcomes.

## Era 2 — Conversation (2023)

ChatGPT, Claude. Turn-based. You ask, it answers.
You are still the executor — you copy-paste, you run, you decide.

## Era 3 — Agency (2024–now)

The model reads your repo, creates a branch, writes tests, runs them, reads the failure, tries again, and opens a PR. You review the PR. You are no longer the executor.

JS COMMUNITY

# LLMs VS Coding Agents

## LLM

User prompt

↓

**LLM**
Predicts next token

↓

Text response

### Characteristics

- Single inference pass
- No memory across turns
- No tool access
- Stateless
- Answers from training data
- Best for: Q&A, drafting, explain

## Coding Agent

User task

↓

**LLM (reasoning)**
Plans + generates actions

↓

Read file | Run code | Write file

↻ *observe → decide → act loop*

↓

Working code / artifact

### Characteristics

- Multi-step agentic loop
- Uses tools (files, shell, APIs)
- Maintains state across steps

JS COMMUNITY

**LLM** : Single forward pass. No memory, no tools, no state. **Coding Agents** : Looping agent with tools, memory, and action feedback!

# Anatomy of an Agent

**Perception → Plan → Act → Remember**

## Perceive

Receive input — a user message, a tool result, a file read. Everything the agent knows about the current task lives in the context window.

1

## Plan

Decide what to do next via chain-of-thought reasoning. Modern **LLMs** do this implicitly. Some frameworks make it explicit with a separate planner model.

2

## Act

Call a tool, write a file, send a message, spawn a sub-agent. This is where real-world effects happen. **Tool** use is the upgrade from chatbot to agent.

3

## Remember

**Store state** for the next iteration — in-context, episodic (vector store), or procedural (structured state object). Context windows are the new heap.

4

**loop**

# Context Engineering

## What is Context?

Everything in the model's window: system prompt, conversation history, tool results, injected docs.

## Compaction

Summarize old turns, prune stale data, keep recent events verbatim. Prevents context overflow and cost blowout.

## Context Rot

Gradual degradation from irrelevant history. Symptoms: hallucinations, repetition, missed instructions.

## Context Window

### STAY ZONE

Recent events, active task, current file

*Always keep verbatim*

### COMPRESS

Older turns, completed sub-tasks

*Summarize before overflow*

### PRUNE

Stale context, resolved errors, old docs

*Remove to fight context rot*

JS COMMUNITY

# Context Engineering

## Context Window

200,000 Token Limit

**System Prompt**
Gives the LLM an identity (i.e. "you are Jarvis")

**Current User Prompt**
i.e. Fix my failing unit tests

**Instructions**
i.e. CLAUDE.md, .github/copilot-instructions.md

**Available Tools**
i.e. Bash, github_create_pr, get_jira_item, etc.

**Sub-Agent Names**
i.e. .claude/agents, .github/agents

**Skill Names**
i.e. .claude/skills, .github/skills

**Plans and TODO lists**
Strategic planning & execution roadmap

---

**Previous Tool Responses**
i.e. code read, PRs created, test results

**Chat History**
Conversation context & decisions made

↓ Free Space ↓

## Dynamically Loaded

When Needed

**Agents**

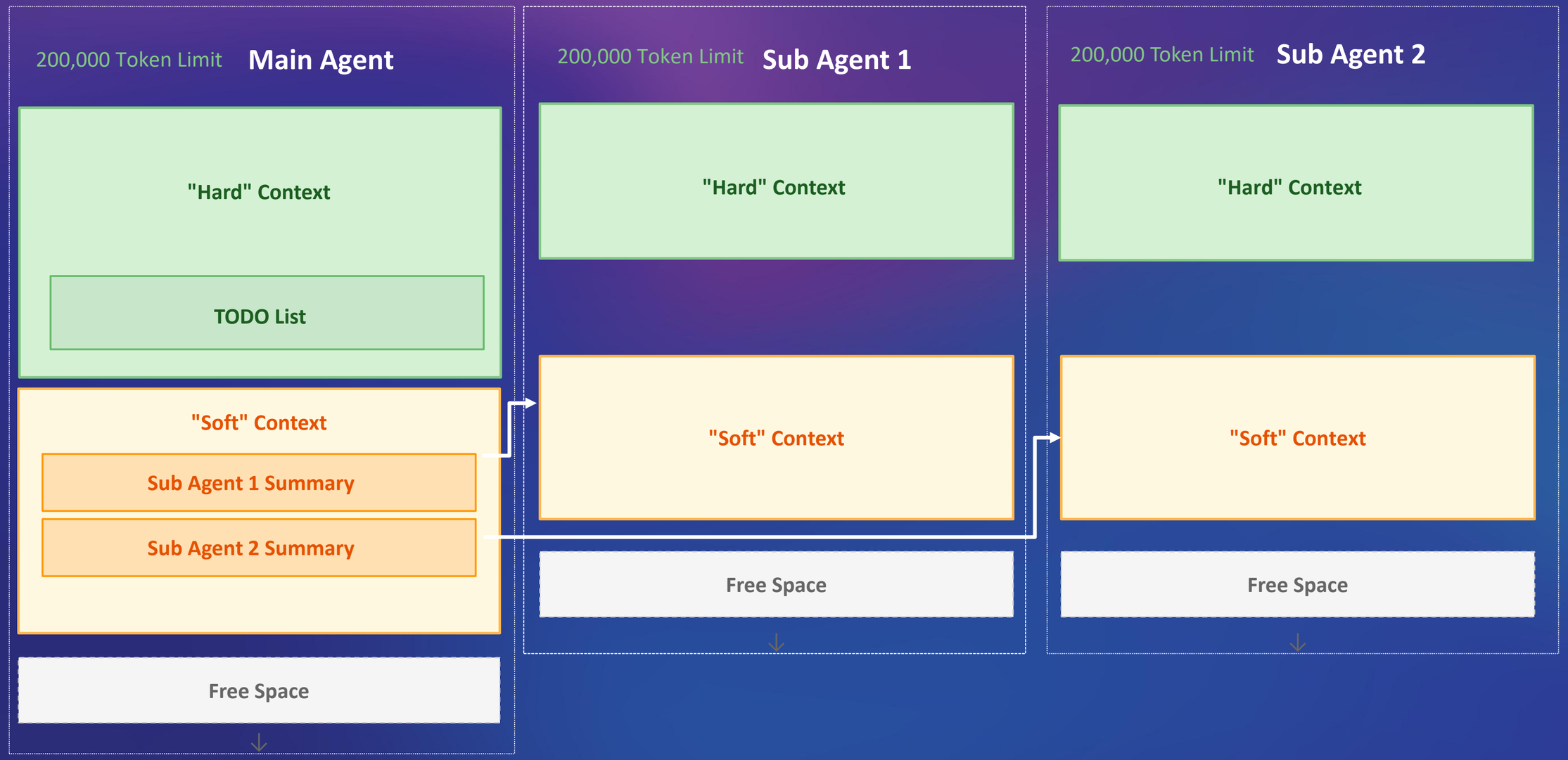
- Agent 1
- Agent 2
- . . .

**Skills**

- Skill 1
- Skill 2
- . . .

Context engineering is the highest-leverage skill in agentic AI development. Master the window, master the agent!

# Sub-agents




Agents can have unique roles with unique perspectives!

# Agent Skills (agentskills.io)

## Agent

200,000 Token Limit

**"Hard" Context**

**Available Skills:**
- xlsx
- ppt
- webapp-testing

**SKILL.md**
scripts/
references/
assets/

**"Soft" Context**

**Free Space**

↓

## Dynamically Loaded

When Needed

**"xlsx" Skill**

**"ppt" Skill**

**"webapp-testing" Skill**

**"c4-diagrams" Skill**

Agents only load the skills they need, maximizing available context space!

# Anatomy of a Skill

```
my-skill/
  ├── SKILL.md
  |   ├── YAML frontmatter
  |   |   (name, description)
  |   └── Markdown instructions
  ├── scripts/
  |   └── Reusable code
  ├── references/
  |   └── Docs loaded on demand
  └── assets/
      └── Templates, fonts, icons
```

**1 Metadata**

Name + description always in context. ~100 words. This is the trigger.

**2 SKILL.md Body**

Full instructions loaded when the skill triggers. Keep under 500 lines.

**3 Bundled Resources**

Scripts, references, assets loaded as needed. Unlimited size.

A skill is a mini-playbook that tells your agent exactly how to think and act!

JS COMMUNITY

# My Creation : The Accessibility Guardian

The Agent: A11y-Healer
File: plugin/a11y-healer/agents/a11y-healer.md

An interactive agent that operates in action mode — it doesn't just suggest changes; it applies them after user confirmation. Workflow:

Explain what it will change and why
Wait for approval
Execute the edit via tools
Verify the change was applied
Run tests to confirm the fix works
Standards enforced: WCAG 2.2 AA, ARIA 1.2, Section 508, ADA.

The agent has an embedded WCAG quick reference covering keyboard patterns (Tab/Shift Tab, arrow keys, Esc, focus order, skip links) and ARIA semantics.

**A11y-Healer Agent**
├── **Skill: accessibility-rule-detection** — WCAG 2.2 violation detection across all POUR principles
│ ├── Detects: missing alt text, low contrast, missing labels
│ ├── Detects: keyboard traps, focus issues
│ └── Detects: improper heading hierarchy, misuse of roles
│
├── **Skill: aria-fix-generation** — Accessible names, ARIA roles/states, error associations, live regions
│ ├── Adds aria-label / aria-labelledby
│ ├── Fixes role attributes and states
│ └── Associates form inputs with error messages
│
├── **Skill: semantic-html-correction** — Native HTML elements, heading hierarchy, landmarks, contrast, focus indicators
│ ├── Replaces div/span with semantic elements (button, nav, header)
│ ├── Fixes heading order (h1 → h6)
│ └── Ensures proper landmarks and focus visibility
│
└── **Skill: accessibility-testing** — axe-core + Playwright/Jest/Cypress, CLI tools, CI/CD integration, manual testing
├── Runs automated accessibility scans
├── Generates accessibility reports
└── Validates fixes before merge in CI/CD

JS COMMUNITY

# Tweak the Prompt, Not the Code

Your prompt IS your program. Quality in → quality out.

❌ BAD

**Bad Prompt**

**"Write code for my app"**

*No context · No role · No constraints · No goal*

→

**LLM**

processes context

→

**Bad Output**

"Generic boilerplate, wrong framework, no tests"

*Useless · Needs complete rewrite*

✅ GOOD

**Good Prompt**

**"You are a senior TS dev. Add auth to this Express route..."**

*Role · Context · Constraints · Goal*

→

**LLM**

same model, same size

→

**Great Output**

"JWT middleware, error handling, tests, typed, production-ready"

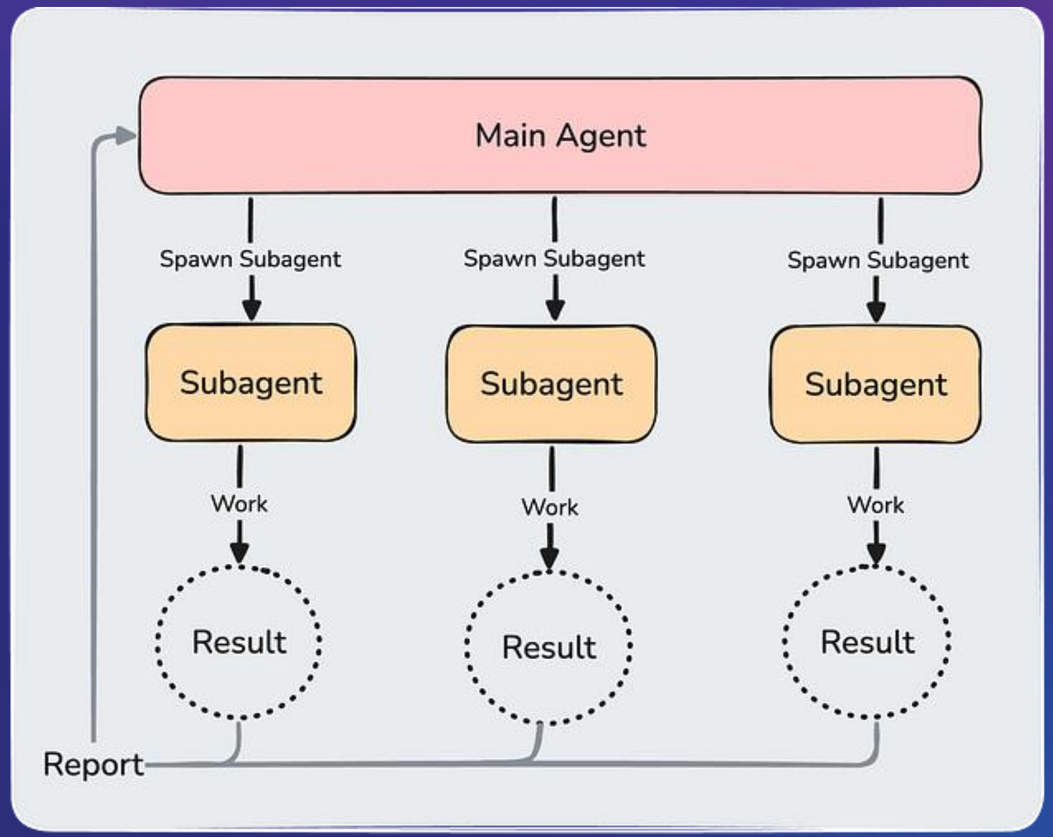
*Ready to ship · Zero rewrite needed*

JS COMMUNITY

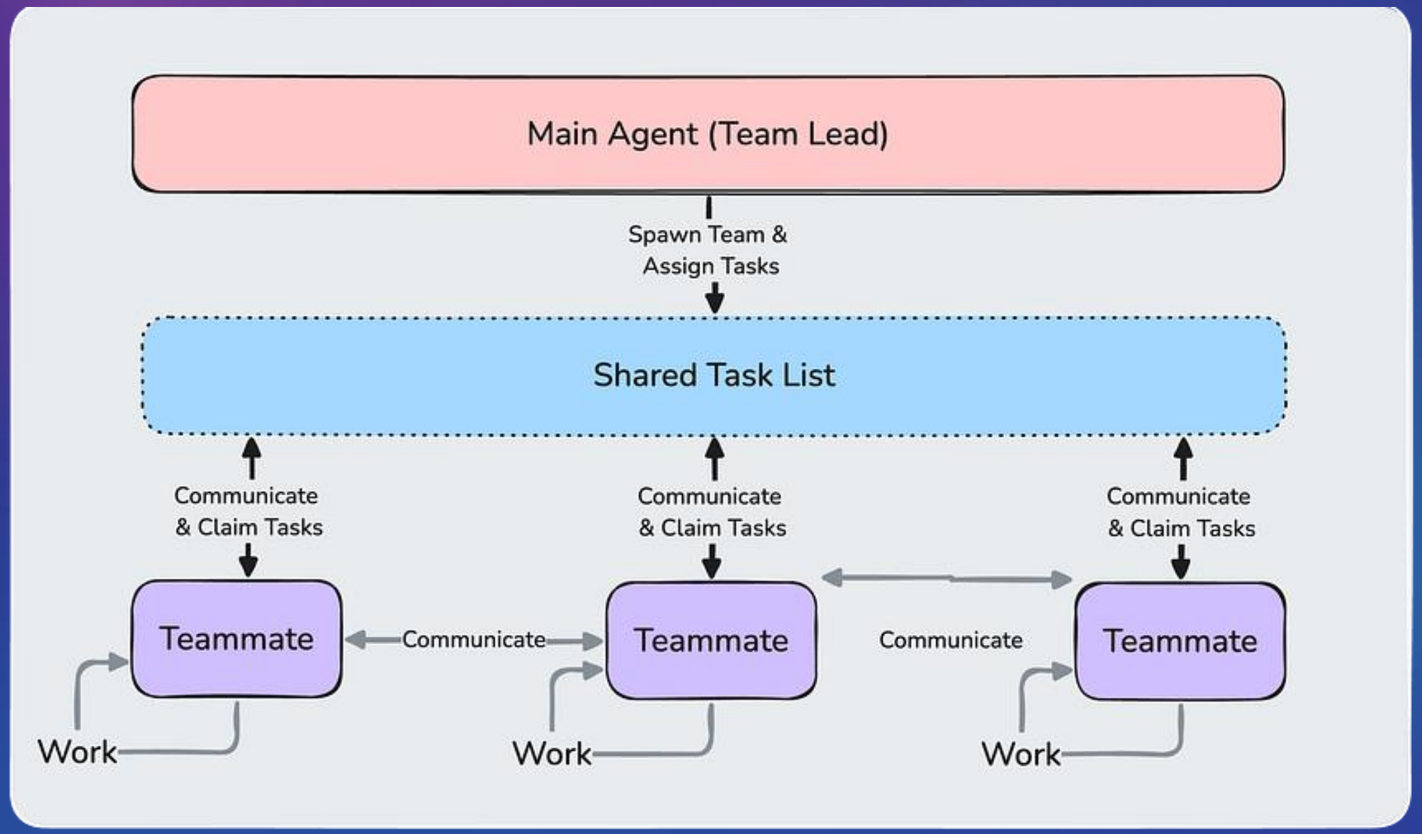
The model didn't change. The prompt did. That's the whole lesson

# Claude Agent Teams

## Subagents




## Agent Teams




Credits : YouTuber Link

# It's Showtime

*Three specialist agents. One codebase. Running in parallel*

## Iron Man

**Security Auditor**

*src/routes/*

- SQL injection
- missing auth
- data exposure

## Captain America

**Auth & Validation**

*src/middleware/*

- Hardcoded secrets
- weak JWT
- no rate limit

## Thor

**Data Integrity**

*src/models/*

- Plaintext passwords
- default admin
- no pagination

*All three run simultaneously · Promise.all([ironMan(), captainAmerica(), thor()])*

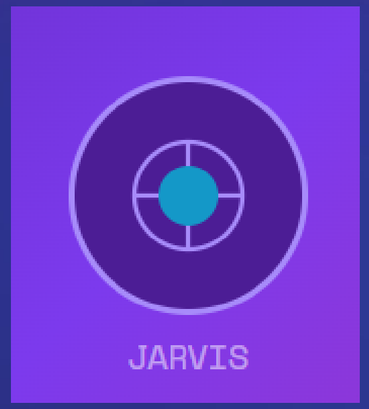
**Mission: find every security, auth and data issue in this codebase — before it ships**

More at :
https://github.com/FlorianBruniaux/claude-code-ultimate-guide/blob/main/guide/workflows/agent-teams.md

Agents in the JS Ecosystem

Teach It Once, Ship It Right

# Build JARVIS, Not Ultron




```
async function runWithCheckpoint(fn) {
  const result = await fn();
  if (result.confidence < 0.75) {
    await requestHumanApproval(result);
  }
  return result;
}
```

## Trust boundaries

Write down what this agent is allowed to do without asking — before you build it. Scope permissions tightly. This is the most important design decision you will make.

## Checkpoint gates

After each agent completes, require a human approval token before the next starts. Put humans in the loop at the right moments — not every step, but the high-stakes ones.

## Audit logs

Every tool call, decision, and retry must be logged in a structured, queryable format. You need to reconstruct exactly what the agent did and why — especially when it goes wrong.

## Async agents ahead

Agents that run in the background, wake on events, persist memory across sessions, and negotiate with each other. The engineer's new role: architect of the system, not executor of the steps.

JS COMMUNITY

# You are the TONY STARK

**You define the mission:**
Frame the problem, define constraints, set trust boundaries. That is the strategic layer — and it stays human.

**JARVIS does the work:**
Review code, generate tests, run CI, report to Jira. Agents handle the repetitive high-stakes execution — without you in the loop for every step.

**You review, not execute:**
Your job shifted from "do the thing" to "approve or redirect the thing." Automated evaluation and human review form a tight loop that improves quality while preserving control.

**Monday morning:**
Pick one repetitive task, prototype an agent for it. Add structured logging to every AI call you're already making.

The engineer's role is not disappearing — it is being promoted

# Thank You

https://github.com/Ashadeepa/ashadeepa-tech-talks

https://www.linkedin.com/in/ashadeepa-debnath/

JS COMMUNITY

And Welcome to the New Agentic Era!